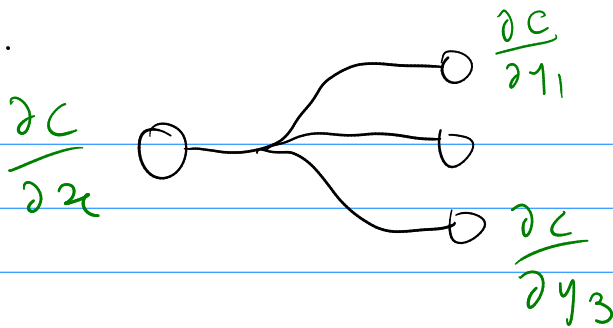
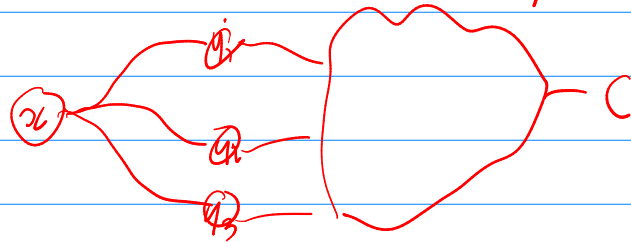


$$\frac{\partial C}{\partial x} = \sum_i \frac{\partial C}{\partial y_i}$$

$$\delta C = \frac{\partial C}{\partial y_1} \delta y_1 + \frac{\partial C}{\partial y_2} \delta y_2 + \frac{\partial C}{\partial y_3} \delta y_3$$



$$\delta C = \delta x \left[\frac{\partial C}{\partial y_1} + \frac{\partial C}{\partial y_2} + \frac{\partial C}{\partial y_3} \right]$$

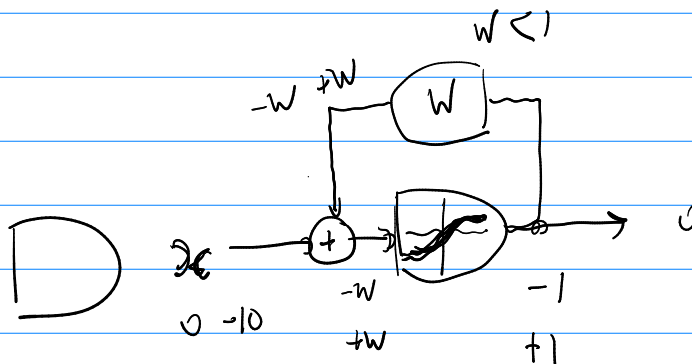
$$\frac{\partial C}{\partial x} = \sum_i \frac{\partial C}{\partial y_i}$$

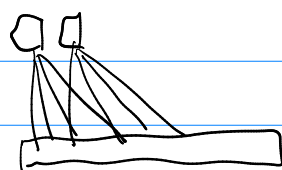
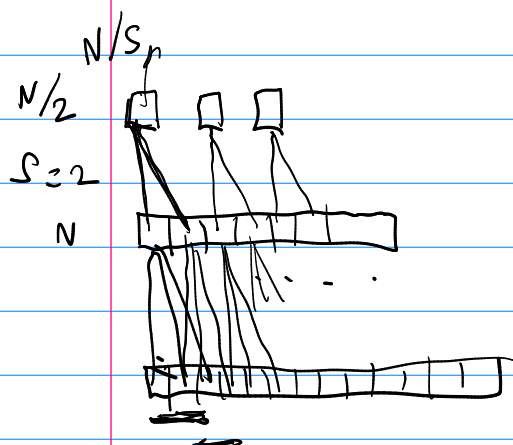
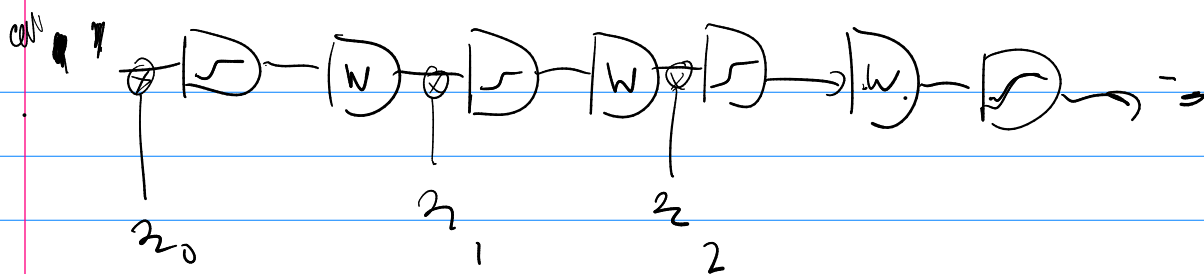
$$z_t \quad \boxed{W} \quad z_{t+1} = W z_t + x_t$$

$$\begin{pmatrix} z_0 \\ z_1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} z_0 \\ z_1 \end{pmatrix} \quad x_t \quad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \quad \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Memory





Pooling : aggregation values
permutation invariant

$$f(x_1, x_2, \dots, x_k) = f(x_{p(1)}, x_{p(2)}, \dots, x_{p(m)})$$

$$\rightarrow y = \frac{1}{k} \sum_{k=1}^k x_k \quad \text{average pooling}$$

$$\rightarrow y = \left(\sum_k x_k^p \right)^{1/p} \quad L_p \text{ pooling}$$

$$\rightarrow y = \max_k (x_k) \quad \text{max pooling}$$

$$\rightarrow y = \frac{1}{p} \log \sum_k e^{\beta x_k} \quad \text{log sum exp pooling}$$

$$f \text{ invariant to } T \quad f(T(x)) = f(x) \quad T(x)_i = x_{i+T}$$

$$f \text{ equivariant to } T \quad f(T(x)) = T'(f(x)) \quad x_i$$

$$\text{Convolution equivariant to translation} \quad f(T(x)) = T(f(x))$$

$$\text{pooling locally invariant to translation} \quad \uparrow \quad \uparrow$$

